Hydrogen Purification System

Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Hydrogen purification systems with alert labels at important parts

What it is used for

To remove specified impurities from hydrogen gas using an approved hydrogen purification system. To maintain required gas quality while controlling pressure, monitoring alerts, and preventing leaks or fire hazards.

Main parts

- Hydrogen inlet and outlet connections
- Purification module or filter bed
- Pressure regulators and gauges
- Flow meter and control valves
- Gas-quality sensors and monitoring display
- Vent line, emergency shutoff, and alert labels

Before you start

1. Only trained and authorized personnel may operate the system.
2. Review the system manual, process flow diagram, safety data, and emergency procedures.
3. Verify that alert labels, flow-direction markings, and hazard signs are visible and legible.
4. Inspect tubing, fittings, seals, valves, and the enclosure for damage or signs of leakage.
5. Confirm that ventilation, gas detection, emergency shutoff, and required safety systems are functional.
6. Check that the inlet hydrogen supply and downstream equipment meet the system's pressure and purity requirements.
7. Verify that the vent outlet discharges to the approved safe location and is not blocked.

How to use

1. Confirm that the system is correctly installed and all required inspections are complete.
2. Keep ignition sources away and ensure the designated operating area is controlled.
3. Verify that downstream equipment is ready and the required outlet path is open.
4. Open the approved supply isolation valve according to the manufacturer's startup procedure.
5. Increase pressure and flow gradually within the specified operating limits.
6. Monitor inlet and outlet pressure, flow, temperature, and gas-quality readings as applicable.
7. Check system indicators and alarms throughout operation.
8. Confirm that purified hydrogen meets the specified quality criteria before sending it to the process.
9. Record operating readings, alarm events, and any required gas-quality test results.

10. If an alarm or abnormal reading occurs, follow the approved response procedure and isolate the system when safe.
11. At shutdown, stop the process and isolate the hydrogen supply in the sequence specified by the manufacturer.
12. Depressurize and purge only according to the approved procedure and designated vent arrangement.
13. Verify the system is in its required safe state before maintenance or opening any component.

⚠ Safety

- Hydrogen is highly flammable and can form explosive mixtures with air; eliminate ignition sources and follow the site's hydrogen safety rules.
- Never use flames to locate leaks; use approved hydrogen leak-detection methods.
- Do not bypass gas detectors, pressure relief devices, alarms, interlocks, or emergency shutoffs.
- Never exceed the rated pressure, temperature, flow, or gas-purity limits.
- Do not loosen fittings or open filters while the system is pressurized or contains hydrogen.
- Use only approved, compatible components and maintain suitable ventilation and hazardous-area controls.
- If a hydrogen leak is suspected, raise the alarm, avoid operating electrical switches in the affected area, evacuate as required, and follow the site emergency plan.

Care & storage

- Inspect labels, detectors, valves, fittings, and seals at the scheduled intervals.
- Replace purification media and filters according to the manufacturer's service instructions.
- Calibrate gas-quality sensors and pressure instruments as required.
- Keep the enclosure and ventilation paths clear of dust and obstructions.
- Document inspections, maintenance, alarms, and any replaced components.

Common mistakes to avoid

- Starting the system without checking ventilation, alarms, and leak-detection readiness.
- Opening valves too quickly and causing pressure surges.
- Ignoring gas-quality readings or repeated alarms.
- Using incompatible seals or unapproved replacement filters.
- Opening a pressurized component or attempting an improvised leak repair.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.